(313)-246-3308
sainalam@umich.edu

Yogendra Sai Pavan Nalam

Applications Programmer Associate

linkedin.com/in/Yogendra Sai Pavan Nalam

COVER LETTER

Date: October 24, 2025

Hiring Committee,
University of Michigan

Dear Hiring Manager,

I am writing to express my interest in the Applications Programmer Associate position at the University of Michigan. As a current Master's in Data Science student at the University of Michigan–Dearborn with hands-on experience in application development, data engineering, and systems integration, I am excited by the opportunity to contribute to the university's mission of innovation and excellence.

In my recent role as a Data Science Intern at Quantum Pulse Consulting, I engineered REST API integrations to embed Power BI dashboards on client websites, improving data accessibility and reducing retrieval bottlenecks by 20%. I also developed machine learning models with 90% accuracy and built RAG pipelines using LLMs to enhance document comprehension experiences that strengthened my full-stack programming, problem-solving, and collaborative skills. As a Graduate Teaching Assistant, I implemented Python/SQL data validation protocols that increased data-quality scores by 40% and led weekly peer reviews, improving software architecture quality for 131 students.

My technical toolkit includes Python, SQL, JavaScript, REST APIs, and cloud platforms like AWS, GCP, and Azure, which aligns closely with the requirements of this role. I am passionate about creating clean, scalable, and secure applications, and I bring a deep familiarity with the University of Michigan environment as both a student and a teaching assistant. This gives me a unique perspective on building tools that directly enhance the campus experience for students, staff, and faculty.

I am excited about the opportunity to grow with the University of Michigan and contribute to projects that drive impactful change. Thank you for considering my application. I look forward to the possibility of discussing how my skills and experience can support your team's goals.

Yours Faithfully
Yogendra Sai Pavan Nalam
(313)-246-3308
sainalam@umich.edu

Yogendra Sai Pavan Nalam

Dearborn, MI | yogendra.nalam@gmail.com | (313)-246-3308 | https://www.linkedin.com/in/yogendra-sai-pavan-nalam/

EDUCATION

University of Michigan - Dearborn Dearborn, MI
Master of Science in Data Science, **GPA: 3.51/4.0** August 2023 - April 2025
Relevant Coursework: Database Systems, Intelligent Systems, Big Data Analytics and Visualization, Artificial Intelligence, Models of Oper Research, Data Mining, Pat Rec & Neural Networks, Applied Regression Analysis, Applied Data Analytics and Modeling for Enterprise Systems, Intro to Big Data.

Vellore Institute of Technology Vellore, India
B.Tech in Computer Science and Engineering, SPL in Bioinformatics, **GPA: 3.40/4.0** July 2019 – May 2023

EXPERIENCE

University of Michigan-Dearborn Michigan, USA
Graduate Teaching Assistant September 2024 - December 2024

- Implemented data-driven feedback loops for 131 students across senior design, using performance metrics to personalize guidance on architectural patterns and software design, leading to tangible improvements in project quality.
- Led weekly peer reviews, enhancing code quality and SOLID adherence, boosting scores by 25 points.
- Implemented Python/SQL data-validation protocols with statistical anomaly detection and integrity checks, boosting data-quality scores by 40% and preventing 16 data-corruption incidents per week.

Quantum Pulse Consulting Michigan, USA
Data Science Intern June 2024 - December 2024

- Engineered seamless client data access by embedding Power BI dashboards onto client websites via REST API integration, reducing data retrieval bottlenecks by 20%.
- Integrated the dashboard with a REST API, enabling seamless access to analytics via the client's website.
- Designed and implemented a multi-dashboard website for 100+ users, boosting workflow efficiency by 40%.
- Implemented a machine learning model with 90% accuracy for financial market analysis, enhancing strategic decision-making.
- Built a Retrieval-Augmented Generation (RAG) pipeline using LLMs to enhance document comprehension.

University of Michigan-Dearborn Michigan, USA
Teaching Assistant June 2024 - July 2024

- Conducted weekly lab sessions for 33 students, teaching coding in HTML, CSS, JavaScript, Node.js, Express, and Vue.js.
- Led 16 lab activities to enhance coding skills and guide students in building interactive, user-friendly web pages.

SKILLS

Data Analysis Techniques: Regression Analysis, Big Data Analytics, Data Mining, Predictive Modeling, Data Cleaning and Wrangling (Pandas, Numpy), Data Visualization, Statistical Analysis, Machine Learning, Generative AI, Prompt Engineering, n8n, Model Context Protocol (MCP)
Programming Languages: Python (Proficient), R (Proficient), SQL (Proficient), C, C++, Java, Shell Scripting
Software and Tools: AWS, GCP, Oracle Cloud, Azure, SAP, REST APIs, Advanced Excel, GitHub, Tableau, Power BI, Databricks, MS Excel, Caffe, PySpark, Microsoft Fabric

CERTIFICATIONS

Microsoft Certified: Fabric Data Engineer Associate(DP 700)
Microsoft Power BI Data Analyst Professional Certificate(PL 300)
Microsoft AI & ML Engineering Professional Certificate

ACADEMIC PROJECTS

Real-Time Traffic Sign Recognition Using YOLO v11 and CNN September 2024 - December 2024

- Achieved 98.7% top-1 accuracy using YOLO v11 on the GTSDB dataset of 39,209 images with optimized data augmentation.
- Designed and trained a custom CNN with TensorFlow for classifying 43 traffic sign classes.
- Conducted comparative analysis with confusion matrices to evaluate model accuracy and misclassification rates.
- Demonstrated YOLO v11's suitability for real-time traffic sign recognition applications.

Classification of Distributed Denial of Service Attacks Using ML Models August 2023 - December 2023

- Identified critical indicators, including packet count, protocol usage, duration, and source IP, enhancing prediction accuracy of malicious DDoS attacks by 45% and reducing false positives by 30%.
- Fortified network security by engineering Random Forest models, reaching 99.8% accuracy in distinguishing malicious attacks from benign traffic, and diminished potential security breaches by 60%.
- Automated a data pipeline using Apache Airflow, which integrated and preprocessed network traffic data, leading to a 25% reduction in data preparation time for security analysts on the team.

Heart Disease Prediction Using Machine Learning January 2023 - April 2023

- Created a real-time heart disease prediction system with 85.8% accuracy using Logistic Regression, Random Forest, and KNN on patient data from Sri Ramachandra Hospital.
- Analyzed a dataset with 12 attributes to identify critical risk factors for early heart disease diagnosis.
- Enhanced model efficiency via advanced attribute selection, improving heart disease diagnosis and patient care.